WORKED SOLUTION

STEP I 1994, Question 6

Pure • Complex Numbers

STEP I 1994, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q6

The key idea

To find the real and imaginary parts of a quotient, multiply top and bottom by the conjugate of the denominator; with x real, that conjugate is easy to write down. For the repeated function, it is much cleaner to compose f with itself *algebraically*, for a general z, than to push the messy real and imaginary parts from (i) through the formula a second time. Once $\mathrm{f}(\mathrm{f}(z))$ is in a tidy form, the last part is a one-line substitution.

Solution

Since x is real, $\mathrm{i}x+1$ has real part 1, so it is never zero and $\mathrm{f}(x)$ is defined for every real x.

Part (i)

The conjugate of $\mathrm{i}x+1$ is $1-\mathrm{i}x$, because x is real. Multiplying top and bottom by it,

$$\begin{aligned}\mathrm{f}(x) = \frac{(\mathrm{i}x-1)(1-\mathrm{i}x)}{(1+\mathrm{i}x)(1-\mathrm{i}x)} &= \frac{\mathrm{i}x+x^2-1+\mathrm{i}x}{1+x^2} \\ &= \frac{x^2-1}{x^2+1} + \mathrm{i}\,\frac{2x}{x^2+1}.\end{aligned}$$

Both fractions are real, so these are the real and imaginary parts, as required.

Part (ii)

For any complex w, $ww^* = (\mathrm{Re}\,w)^2 + (\mathrm{Im}\,w)^2$. Using part (i),

$$\mathrm{f}(x)\mathrm{f}(x)^* = \frac{(x^2-1)^2+4x^2}{(x^2+1)^2} = \frac{x^4+2x^2+1}{(x^2+1)^2} = 1.$$

So $|\mathrm{f}(x)| = 1$: f sends every real number to a point on the unit circle.

Part (iii)

Rather than substituting the answer to (i) into f, we compose for a general z. Put $w = \mathrm{f}(z) = \dfrac{\mathrm{i}z - 1}{\mathrm{i}z + 1}$ and multiply the top and bottom of $\mathrm{f}(w) = \dfrac{\mathrm{i}w - 1}{\mathrm{i}w + 1}$ by $\mathrm{i}z + 1$:

$$\begin{aligned}\mathrm{f}(\mathrm{f}(z)) &= \frac{\mathrm{i}(\mathrm{i}z - 1) - (\mathrm{i}z + 1)}{\mathrm{i}(\mathrm{i}z - 1) + (\mathrm{i}z + 1)} = \frac{-z - \mathrm{i} - \mathrm{i}z - 1}{-z - \mathrm{i} + \mathrm{i}z + 1} \\ &= \frac{-(1+\mathrm{i})(z+1)}{(1-\mathrm{i})(1-z)} = \frac{1+\mathrm{i}}{1-\mathrm{i}} \cdot \frac{z+1}{z-1} = \mathrm{i}\,\frac{z+1}{z-1},\end{aligned}$$

since $\dfrac{1+\mathrm{i}}{1-\mathrm{i}} = \dfrac{(1+\mathrm{i})^2}{2} = \mathrm{i}$.

Ask yourself Is $\mathrm{f}(\mathrm{f}(x))$ defined for every real x?

No. We need $\mathrm{i}w + 1 \neq 0$, that is, $\mathrm{f}(x) \neq \mathrm{i}$. From part (i), $\mathrm{f}(x) = \mathrm{i}$ needs $x^2 - 1 = 0$ and $2x = x^2 + 1$, which happens only for $x = 1$. This matches the factor $x - 1$ in the denominator above. So for real $x \neq 1$,

$$\mathrm{f}(\mathrm{f}(x)) = \mathrm{i}\,\frac{x+1}{x-1}, \qquad \text{so} \qquad \operatorname{Re}\mathrm{f}(\mathrm{f}(x)) = 0, \quad \operatorname{Im}\mathrm{f}(\mathrm{f}(x)) = \frac{x+1}{x-1}.$$

The real line is sent to the unit circle by the first application of f, and the unit circle is sent to the imaginary axis by the second.

Part (iv)

Write $\mathrm{f}(\mathrm{f}(x)) = \mathrm{i}y$ with $y = \dfrac{x+1}{x-1}$ real. Then $\mathrm{i}(\mathrm{i}y) = -y$, so

$$\mathrm{f}(\mathrm{i}y) = \frac{-y-1}{-y+1} = \frac{y+1}{y-1}.$$

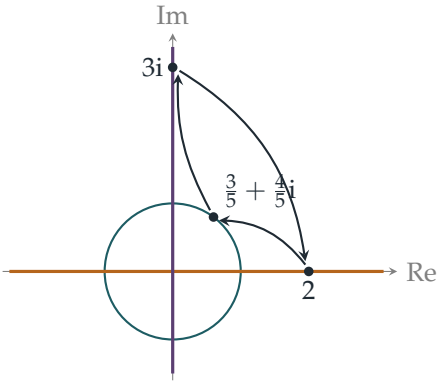
This is defined because $y = 1$ would need $x + 1 = x - 1$, which is impossible. Now

$$y + 1 = \frac{2x}{x-1}, \qquad y - 1 = \frac{2}{x-1},$$

so

$$\mathrm{f}(\mathrm{f}(\mathrm{f}(x))) = x \qquad \text{for every real } x \neq 1.$$

Applying f three times brings us back to where we started: the third application sends the imaginary axis back to the real line. The picture shows the cycle for $x = 2$, which goes $2 \mapsto \frac{3}{5} + \frac{4}{5}\mathrm{i} \mapsto 3\mathrm{i} \mapsto 2$.



Checks

- $x = 0$: $\mathrm{f}(0) = -1$, which is on the unit circle and agrees with (i). Then $\mathrm{f}(-1) = \dfrac{-\mathrm{i}-1}{-\mathrm{i}+1} = -\mathrm{i}$, matching $\mathrm{i}\frac{0+1}{0-1}$, and $\mathrm{f}(-\mathrm{i}) = \frac{1-1}{1+1} = 0$.
- $x = 2$: part (i) gives $\frac{3}{5}+\frac{4}{5}\mathrm{i}$, part (iii) gives 3i, and part (iv) returns 2. A numerical check on 10^4 random real x (and random complex z) confirmed every formula above to rounding error.

Common slips

- Substituting the real and imaginary parts from (i) into f directly. It works, but the algebra is heavy and easy to get wrong. Composing in terms of z is far cleaner.
- Forgetting that $x = 1$ must be excluded from (iii) and (iv), because $\mathrm{f}(1) = \mathrm{i}$ and $\mathrm{f}(\mathrm{i})$ has a zero denominator.

Beyond STEP

A function of the form $z \mapsto \dfrac{az+b}{cz+d}$ is a *Möbius transformation*. Such maps send circles and lines to circles and lines, as we saw here: real axis to unit circle to imaginary axis and back again. Composing two of them corresponds to multiplying the 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. In fact $\mathrm{f}(\mathrm{f}(\mathrm{f}(z))) = z$ for every complex z, so f has order 3 in the group of Möbius transformations. Also, putting $x = \tan\frac{\theta}{2}$ in (i) gives $\mathrm{f}(x) = -\cos\theta + \mathrm{i}\sin\theta$, the familiar "$t$-substitution" parametrisation of the circle. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations and their geometry.

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